

OIBSIP Data Analytics Internship Report

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Task 1 - EDA on Retail Sales

Objective

Perform exploratory data analysis to identify sales trends, customer behaviour and business insights.

Detailed Implementation

The project began by importing the required Python libraries, including Pandas for data manipulation, NumPy for numerical computations, Matplotlib and Seaborn for data visualization. The retail sales dataset was then loaded into a Pandas DataFrame using the `read_csv()` function. The first few rows of the dataset were displayed using `head()` to understand the structure and verify that the data had been imported correctly.

After loading the dataset, an initial data inspection was performed using `info()`, `describe()`, `shape`, `isnull().sum()`, and `duplicated().sum()`. These functions helped identify the number of records, data types, missing values, duplicate records, and summary statistics of the numerical variables. It was observed that the dataset contained very few missing values and no duplicate records, making it suitable for further analysis.

The date columns were converted into the datetime format using Pandas, and additional features such as **Year**, **Month**, and **Quarter** were extracted from the order date. These new variables enabled time-based analysis of sales performance throughout different periods.

Several exploratory data analysis (EDA) techniques were then applied to understand business performance from multiple perspectives. A monthly sales trend line chart was created to visualize seasonal fluctuations in sales. Bar charts were generated to compare sales across customer segments, regions, product categories, and states. A visualization of sales by sub-category and top-selling products was also prepared to identify the best-performing products. Furthermore, a histogram was created to study the distribution of sales values, while a correlation heatmap was used to examine relationships among numerical variables.

Finally, all visualizations were interpreted to extract meaningful business insights. These analyses helped identify the highest-performing customer segment, product categories, geographical regions, and seasonal sales patterns. Based on these observations, practical business recommendations were suggested to improve marketing strategies, inventory planning, and sales performance.

Feature Checklist Mapping

- Dataset loading and initial inspection
- Missing value handling
- Duplicate check
- Data preprocessing
- Visualizations and analysis
- Business insights
- Conclusion

Screenshots

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
1	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase
2	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs
3	CA-2017-138688	12-06-2017	16-06-2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters
4	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table
5	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N' Roll Cart System
6	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Access
7	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10001833	Office Supplies	Art	Newell 322
8	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10001275	Technology	Phones	Mitel 5320 IP Phone VoIP phone
9	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by
10	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C200VTE1 6 Outlet Surge
11	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables
12	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10001033	Technology	Phones	Konftel 250 Conference Phone - Charcoal bla
13	CA-2018-114412	15-04-2018	20-04-2018	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967
14	CA-2017-161380	05-12-2017	10-12-2017	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine
15	US-2016-118983	22-11-2016	26-11-2016	Standard Class	HP-14815	Harold Pavlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Clean
16	US-2016-118983	22-11-2016	26-11-2016	Standard Class	HP-14815	Harold Pavlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10000766	Office Supplies	Binders	Stora DuraTech Recycled Plastic Footed Bnde
17	CA-2015-105893	11-11-2015	18-11-2015	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-10004186	Office Supplies	Storage	Star-D-Stor Shelving, Vertical 5-Shelf, 72"H x 36
18	CA-2015-167164	13-05-2015	15-05-2015	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Drawer
19	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschus Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10000356	Office Supplies	Art	Newell 341
20	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschus Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone
21	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschus Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215	Office Supplies	Binders	Wilson Jones Hanging View Binder, White, 1"
22	CA-2017-137330	09-12-2017	13-12-2017	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10000246	Office Supplies	Art	Newell 318
23	CA-2017-137330	09-12-2017	13-12-2017	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AP-10001492	Office Supplies	Appliances	Acco Six-Outlet Power Strip, 4' Cord Length
24	US-2018-156909	16-07-2018	18-07-2018	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-10002774	Furniture	Chairs	Bretford CR4500 Series Slim Rectangular Table
25	CA-2016-106320	25-09-2016	30-09-2016	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables	Wilson Jones Active Use Binders
26	CA-2017-121755	16-01-2017	20-01-2017	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001634	Office Supplies	Binders	Imation 8GB Mini TravelDrive USB 2.0A Flash
27	CA-2017-121755	16-01-2017	20-01-2017	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10001027	Technology	Accessories	Riverside Palais Royal Lawyers Bookcase, Roya
28	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases	Avery Recycled Flexi-View Covers for Binding S
29	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-10000474	Office Supplies	Binders	Howard Miller 13-3/4" Diameter Brushed Chcon
30	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-FU-10004848	Furniture	Furnishings	Poly String Tie Envelopes
31	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-10001509	Office Supplies	Envelopes	BOSTON Model 1800 Electric Pencil Sharpener
32	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-AR-10004042	Office Supplies	Art	

Figure 1: Dataset Overview

The dataset contains retail order information including customer details, product category, region, sales amount, shipping information, and order dates. Before starting the analysis, the dataset was inspected to understand its structure, identify missing values, verify data types, and ensure the data was suitable for analysis.

	Row ID	Postal Code	Sales
count	9800.000000	9789.000000	9800.000000
mean	4900.500000	55273.322403	230.769059
std	2829.160653	32041.223413	626.651875
min	1.000000	1040.000000	0.444000
25%	2450.750000	23223.000000	17.248000
50%	4900.500000	58103.000000	54.490000
75%	7350.250000	90008.000000	210.605000
max	9800.000000	99301.000000	22638.480000

Figure 2: Descriptive Statistics

The descriptive statistics summarize the numerical variables in the dataset. Measures such as mean, median, minimum, maximum, and standard deviation help understand the overall distribution of sales and identify possible outliers. This provides a strong foundation before performing further analysis.

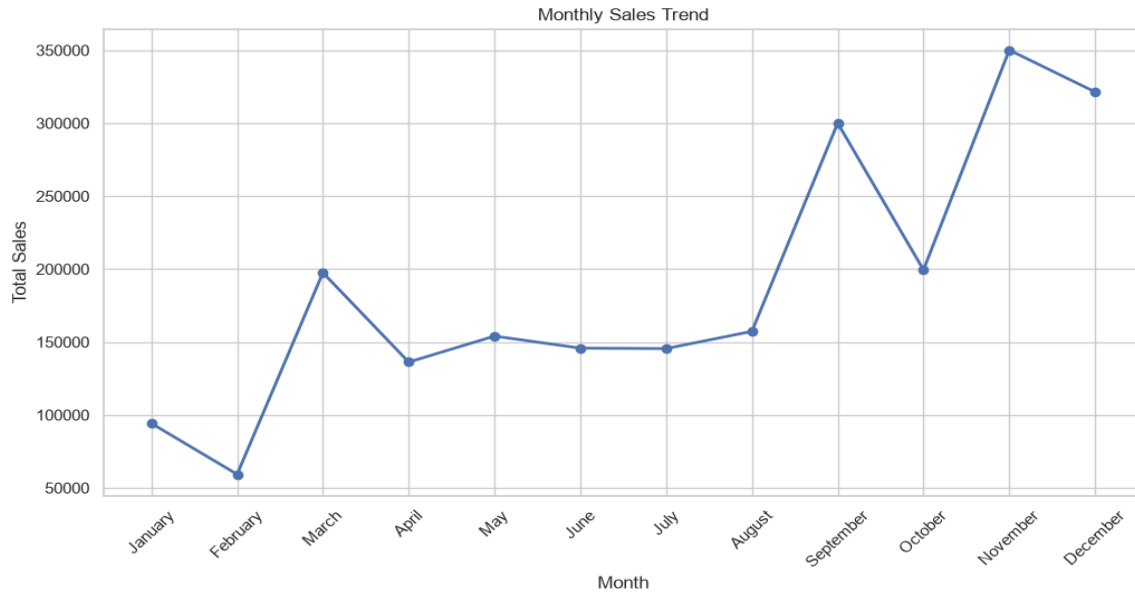


Figure 3: Monthly Sales Trend

The monthly sales trend shows how sales changed throughout the year. Sales remain relatively stable during the middle months but increase significantly during September, November, and December. November records the highest sales, indicating strong seasonal demand and increased customer purchases during the end of the year.

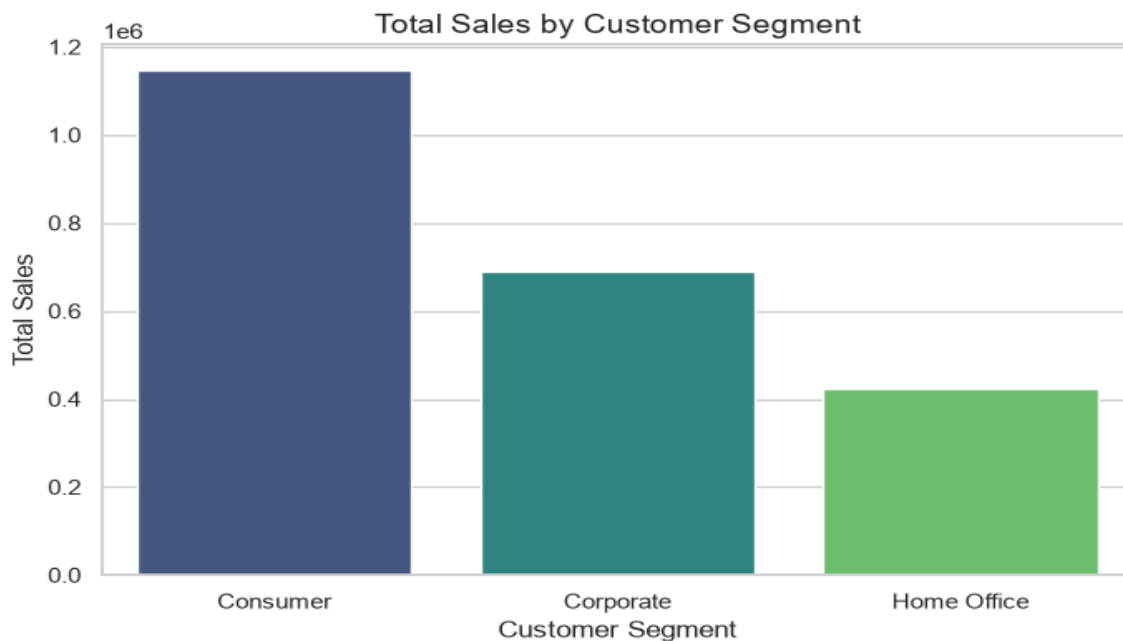


Figure 4: Customer Analysis

The chart compares total sales across customer segments. The Consumer segment contributes the highest sales, followed by Corporate customers, while the Home Office segment generates the lowest revenue. This indicates that individual consumers are the primary source of business revenue.

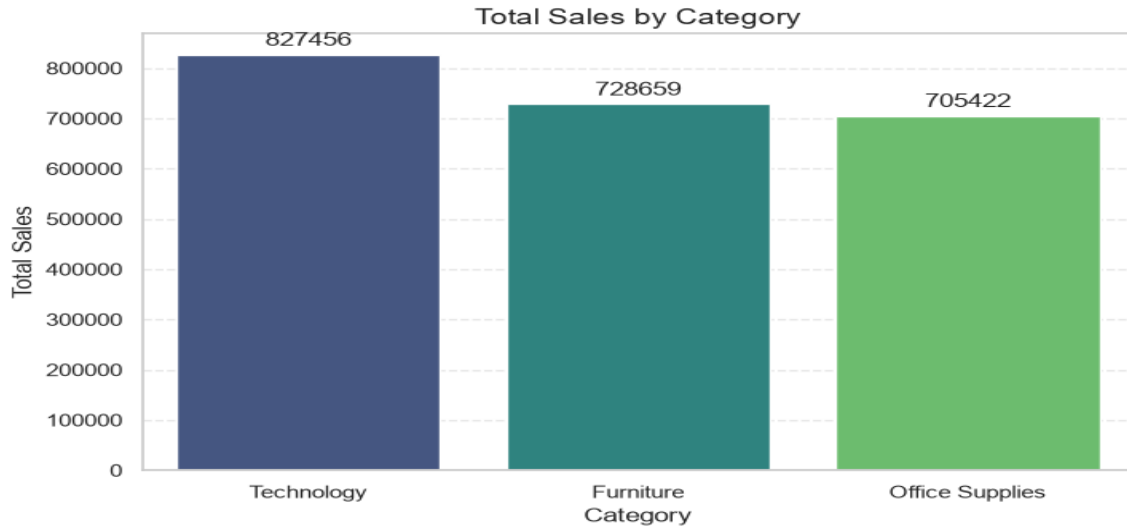


Figure 5: Product Analysis

The category-wise sales chart shows that Technology products generate the highest revenue, followed by Furniture and Office Supplies. Although the difference is not very large, Technology contributes the largest share of overall sales, suggesting strong customer demand.

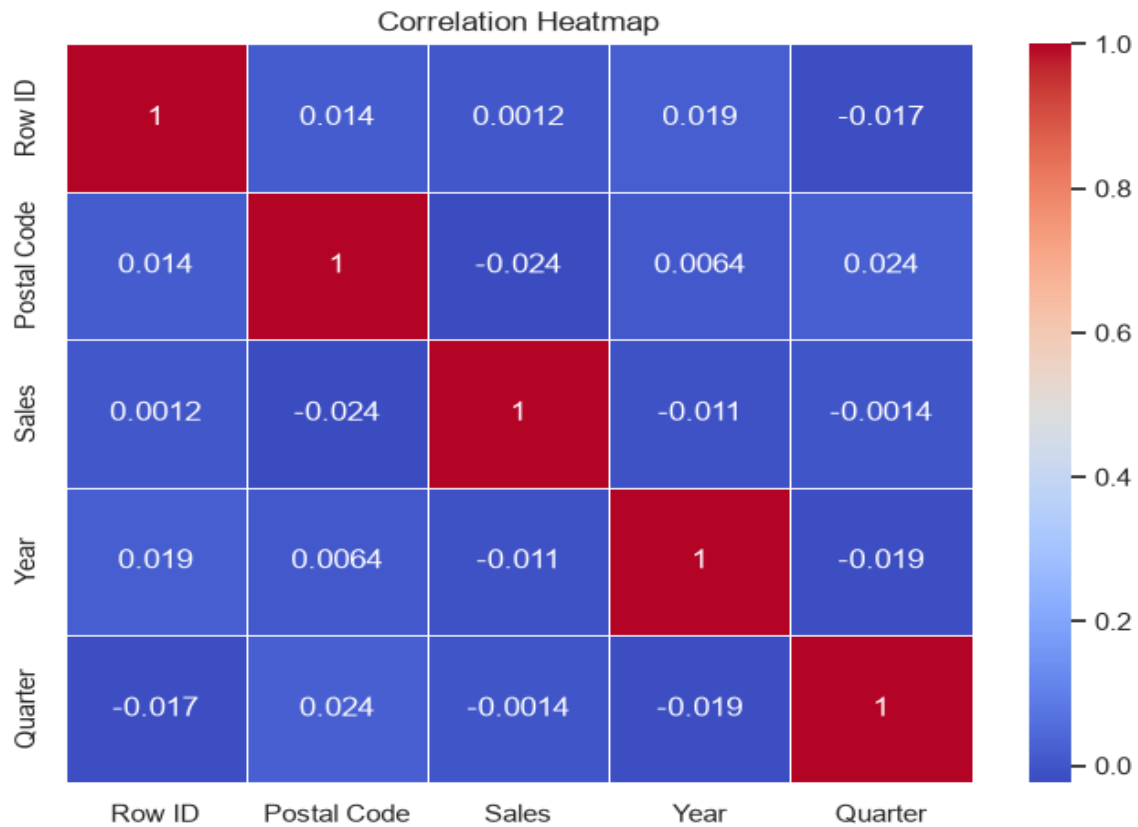


Figure 7: Correlation Heatmap

The correlation heatmap illustrates the relationships between numerical variables. Most correlation values are close to zero, indicating weak linear relationships among the selected variables. No strong correlation is observed with sales, suggesting that sales are influenced by multiple factors rather than a single numerical feature.

Key Findings

- Consumer customers generated the highest sales.
- Technology products were the top-performing category.
- Sales increased significantly during the last quarter.
- California and the West region contributed the highest revenue.
- Sales show seasonal variation across different months.

Business Recommendations

- Increase inventory before the festive season.
- Focus marketing campaigns on Consumer customers.
- Expand Technology product offerings.
- Improve marketing efforts in lower-performing regions.
- Promote underperforming product categories using discounts.

Task Conclusion

The project successfully performed exploratory data analysis on retail sales data. Various visualizations were created to identify customer behaviour, product performance, regional trends, and seasonal sales patterns. The analysis generated meaningful business insights that can support better marketing and inventory planning.

Task 2 - Customer Segmentation using K-Means

Objective

Segment customers into meaningful groups for targeted marketing.

Detailed Implementation

The project started by importing the required libraries, including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. The shopping mall customer dataset was loaded into a Pandas DataFrame, and the first few records were displayed to understand the available customer information.

The dataset was inspected using functions such as `shape`, `info()`, `describe()`, `isnull().sum()`, and `duplicated().sum()`. These checks confirmed that the dataset contained no missing values or duplicate records. The numerical features selected for clustering were **Age**, **Annual Income**, and **Spending Score**, as these variables are highly relevant for customer segmentation.

Before applying the clustering algorithm, feature scaling was performed using the **StandardScaler** class from Scikit-learn. Standardization ensured that all variables contributed equally during distance calculations, preventing features with larger numerical values from dominating the clustering process.

The optimal number of clusters was determined using the **Elbow Method**. The Within-Cluster Sum of Squares (WCSS) was calculated for different values of K, and the resulting graph showed that five clusters provided the best balance between model simplicity and clustering performance.

After selecting the optimal number of clusters, the **K-Means algorithm** was trained using five clusters. Each customer was assigned to one of the clusters based on similarities in age, annual income, and spending score. The cluster labels were then added back to the original dataset.

To visualize the segmentation results, a scatter plot was created using Annual Income and Spending Score as the two axes. Different colors represented different customer clusters, while black "X" markers indicated the cluster centroids. The average age, annual income, and spending score for each cluster were calculated to understand the characteristics of each customer group. Based on these findings, marketing recommendations were provided for targeting different customer segments more effectively.

Feature Checklist Mapping

- Dataset loading and initial inspection
- Missing value handling
- Duplicate check
- Data preprocessing
- Visualizations and analysis

- Business insights
- Conclusion

Screenshots

1	Customer ID	Age	Gender	Annual Income	Spending Score
2	d410ea53-6661-42a9-ad3a-f554b05fd2a7	30	Male	151479	89
3	1770b26f-493f-46b6-837f-4237fb5a314e	58	Female	185088	95
4	e81aa8eb-1767-4b77-87ce-1620dc732c5e	62	Female	70912	76
5	9795712a-ad19-47bf-8886-4f997d6046e3	23	Male	55460	57
6	64139426-2226-4cd6-bf09-91bce4b4db5e	24	Male	153752	76
7	7e211337-e92f-4140-8231-5c9ac7a2aa12	42	Male	158335	40
8	30661d57-eeb5-4886-970a-9c7d0f260ad8	27	Female	163501	37
9	43417ade-6b17-431e-ae76-d8ed26aa46ac	24	Male	70476	17
10	86f73b1a-6a91-4d7e-ad21-0aefcf6111cc	83	Male	47743	34
11	b20fe0a0-e254-4116-b804-5403bb2f59ea	62	Female	63448	3
12	49ba3e9d-a590-4303-b443-e794d0364c11	37	Male	170566	75
13	4be5f5b0-83f6-4bd2-9056-18a8d5414066	63	Female	89355	80
14	eb0e4ebc-fcc1-424f-b083-81987583ecc6	25	Male	87576	17
15	caa4ad5a-51b2-4af1-91cb-e2802f620239	32	Female	159892	96
16	0c9d0e0d-a601-488c-9f3f-f036f2bdb536	51	Female	135079	87
17	dc35a079-c3db-44f4-8612-d0af9daea414	64	Female	32869	81
18	b43f995c-c9ab-4c58-8f71-949de233ac71	19	Male	198682	82
19	51d7be3e-84c9-464a-85e8-0a95b568d90e	89	Male	38708	97
20	8bbdbbbe-7f8a-4c9c-afed-f6969f49e9ac	65	Male	87270	3

Figure 1: Dataset Overview

The customer dataset contains demographic and spending information including age, gender, annual income, and spending score. The data was inspected to ensure consistency before clustering.

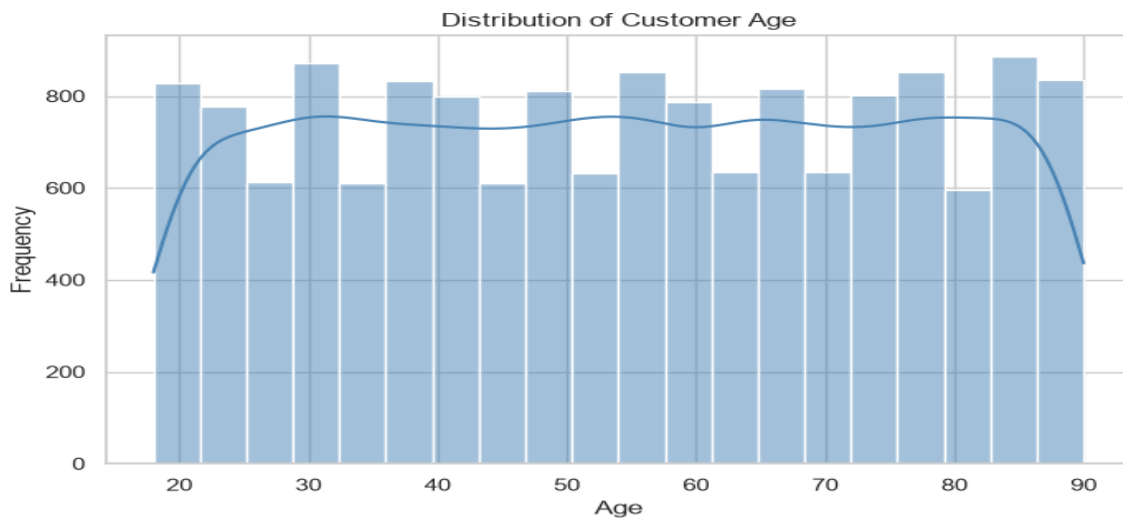


Figure 2: Age Distribution

Customer ages range from approximately 18 to 90 years. The distribution is fairly balanced, indicating that customers belong to different age groups without any significant age concentration.

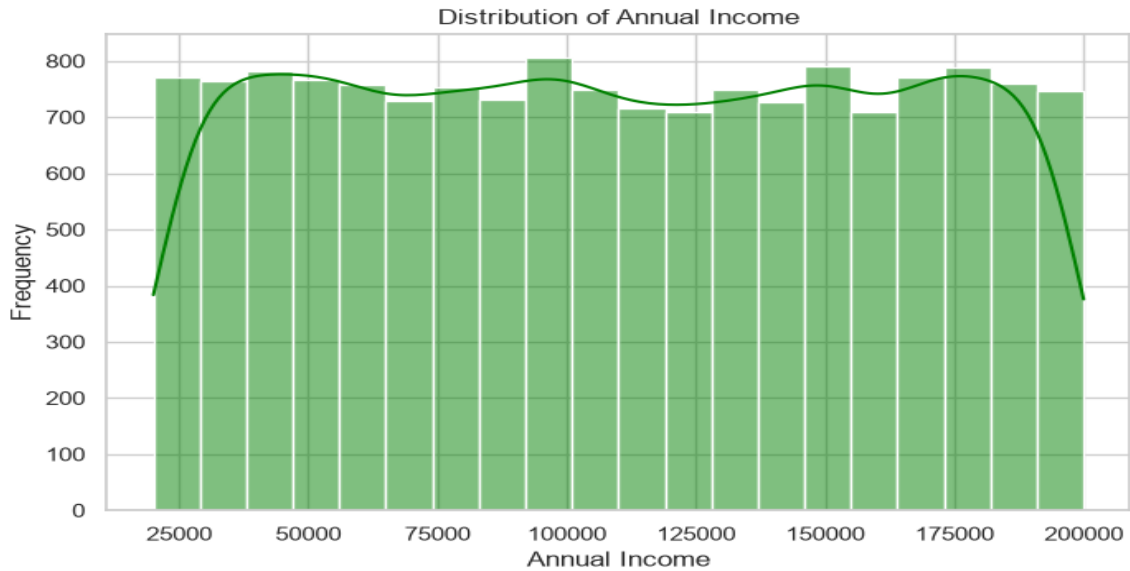


Figure 3: Income Distribution

Annual income is distributed across a wide range, providing sufficient variation for customer segmentation. This feature plays an important role in distinguishing customer purchasing power.

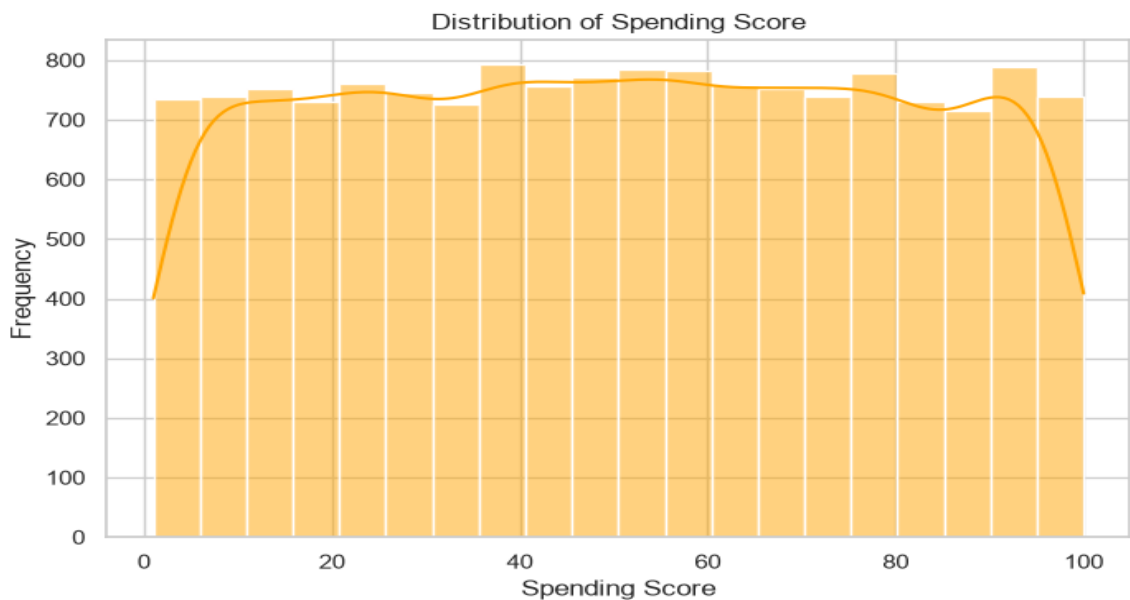


Figure 4: Spending Score Distribution

The spending score ranges from 1 to 100, representing customer purchasing behaviour. Customers are distributed across low, medium, and high spending levels, making it suitable for clustering analysis.

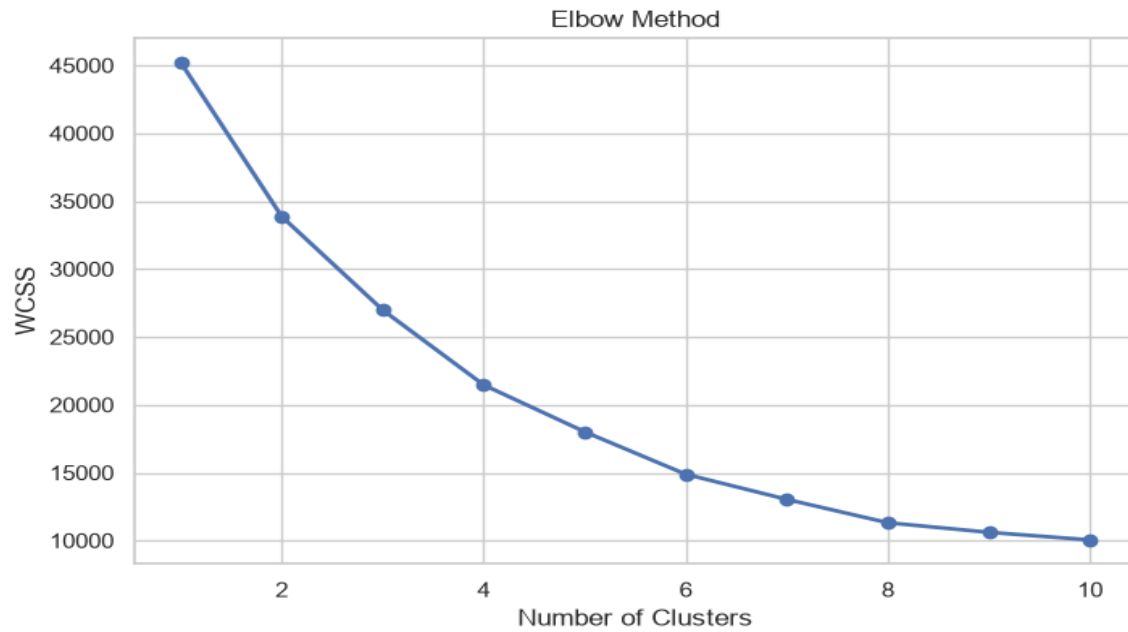


Figure 5: Elbow Method

The elbow method helps determine the optimal number of clusters by plotting WCSS against the number of clusters. The curve begins to flatten around **5 clusters**, indicating that five customer groups provide a good balance between simplicity and clustering performance.

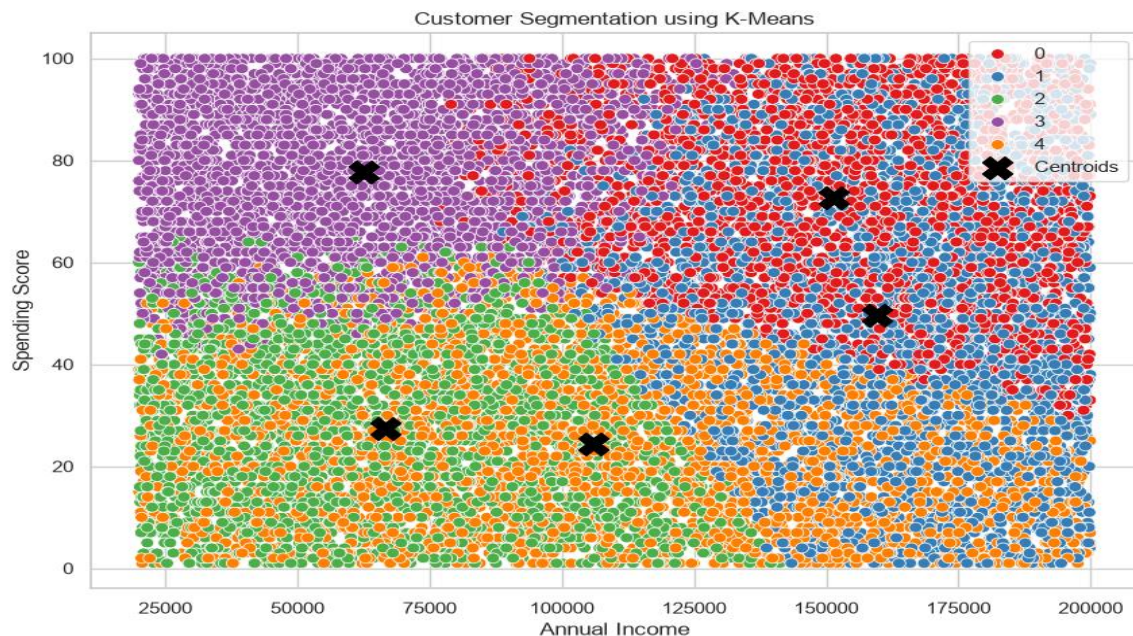


Figure 6: Cluster Plot

The scatter plot displays customers grouped into five clusters based on Annual Income and Spending Score. Each colour represents a unique customer segment, while the black "X" marks the centroid of each cluster.

markers indicate cluster centroids. The clustering successfully separates customers with different purchasing behaviours.

Key Findings

- Five customer clusters were identified.
- High-income customers exhibit different spending behaviours.
- Some customers have high spending despite moderate income.
- Low-income customers generally have lower spending scores.
- Customer segmentation enables targeted marketing strategies.

Business Recommendations

- Offer premium products to high-income, high-spending customers.
- Provide discounts to encourage purchases from low-spending customers.
- Design loyalty programs for frequent buyers.
- Develop personalized marketing campaigns for each customer segment.
- Allocate marketing budgets according to customer value.

Task Conclusion

The K-Means clustering model successfully segmented customers into five distinct groups. These customer segments provide valuable insights that can help businesses improve customer targeting, personalize promotions, and enhance marketing effectiveness.

Task 3 - House Price Prediction

Objective

Build and evaluate a Linear Regression model for house price prediction.

Detailed Implementation

The project began by importing the necessary Python libraries, including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. The Ames Housing dataset was loaded into a Pandas DataFrame, and the initial structure of the dataset was examined using `head()`, `shape`, `info()`, and `describe()`.

A detailed data quality assessment was performed to identify missing values and duplicate records. Several features contained missing values, with some columns having a very high percentage of missing data. Columns containing more than 40% missing values were removed because they contributed limited useful information to the prediction model. The remaining missing numerical values were replaced using the median of their respective columns, while categorical missing values were filled using the most frequently occurring category (mode). Duplicate records were also checked to ensure data consistency.

Since machine learning algorithms require numerical input, all categorical variables were converted into numerical values using **One-Hot Encoding** with the `get_dummies()` function. This transformation increased the number of features while preserving the information contained in categorical variables.

The cleaned dataset was then divided into independent variables (features) and the dependent variable (**SalePrice**). The data was split into training and testing sets using the `train_test_split()` function. A **Linear Regression** model from Scikit-learn was trained using the training dataset and then used to predict house prices for the testing dataset.

The performance of the model was evaluated using several regression metrics, including **Mean Absolute Error (MAE)**, **Mean Squared Error (MSE)**, **Root Mean Squared Error (RMSE)**, and the **R² Score**. These metrics were used to measure the prediction accuracy of the model. Visualizations such as the **Actual vs Predicted** scatter plot and the **Residual Plot** were generated to evaluate prediction performance and identify prediction errors. Additionally, the model coefficients were analyzed to determine which housing features had the greatest influence on sale price. Finally, the overall model performance and business implications were discussed, along with recommendations for improving prediction accuracy in future work.

Feature Checklist Mapping

- Dataset loading and initial inspection
- Missing value handling
- Duplicate check
- Data preprocessing

- Visualizations and analysis
- Business insights
- Conclusion

Screenshots

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Order	PID	MS SubClass	MS Zoning	Lot Frontage	Lot Area	Street	Alley	Lot Shape	Land Contour	Utilities	Lot Config	Land Slope	Neighborhood	Condition 1	Condition 2	Bldg Type	House Style	Overall Qual	Overall Cond	Year Built
2	1	526301100	20	RL	141	31770	Pave	NA	IR1	Lvl	AllPub	Corner	Gtl	NAmes	Norm	Norm	1fam	1story	6	5	1960
3	2	526350040	20	RH	80	11622	Pave	NA	Reg	Lvl	AllPub	Inside	Gtl	NAmes	Feedr	Norm	1fam	1story	5	6	1961
4	3	526351010	20	RL	81	14267	Pave	NA	IR1	Lvl	AllPub	Corner	Gtl	NAmes	Norm	Norm	1fam	1story	6	6	1958
5	4	526353030	20	RL	93	11160	Pave	NA	Reg	Lvl	AllPub	Corner	Gtl	NAmes	Norm	Norm	1fam	1story	7	5	1968
6	5	527105010	60	RL	74	13830	Pave	NA	IR1	Lvl	AllPub	Inside	Gtl	Gilbert	Norm	Norm	1fam	2story	5	5	1997
7	6	527105030	60	RL	78	9978	Pave	NA	IR1	Lvl	AllPub	Inside	Gtl	Gilbert	Norm	Norm	1fam	2story	6	6	1998
8	7	527127150	120	RL	41	4920	Pave	NA	Reg	Lvl	AllPub	Inside	Gtl	StoneBr	Norm	Norm	TwtnhE	1story	8	5	2001
9	8	527145080	120	RL	43	5005	Pave	NA	IR1	HLS	AllPub	Inside	Gtl	StoneBr	Norm	Norm	TwtnhE	1story	8	5	1992
10	9	527146030	120	RL	39	5389	Pave	NA	IR1	Lvl	AllPub	Inside	Gtl	StoneBr	Norm	Norm	TwtnhE	1story	8	5	1995
11	10	527162130	60	RL	60	7500	Pave	NA	Reg	Lvl	AllPub	Inside	Gtl	Gilbert	Norm	Norm	1fam	2story	7	5	1999
12	11	527163010	60	RL	75	10000	Pave	NA	IR1	Lvl	AllPub	Corner	Gtl	Gilbert	Norm	Norm	1fam	2story	6	5	1993
13	12	527165230	20	RL		7980	Pave	NA	IR1	Lvl	AllPub	Inside	Gtl	Gilbert	Norm	Norm	1fam	1story	6	7	1992
14	13	527166040	60	RL	63	8402	Pave	NA	IR1	Lvl	AllPub	Inside	Gtl	Gilbert	Norm	Norm	1fam	2story	6	5	1998
15	14	527180040	20	RL	85	10176	Pave	NA	Reg	Lvl	AllPub	Inside	Gtl	Gilbert	Norm	Norm	1fam	1story	7	5	1990
16	15	527182190	120	RL		6820	Pave	NA	IR1	Lvl	AllPub	Corner	Gtl	StoneBr	Norm	Norm	TwtnhE	1story	8	5	1985
17	16	527214070	60	RL	47	53504	Pave	NA	IR2	HLS	AllPub	CulDSac	Mod	StoneBr	Norm	Norm	1fam	2story	8	5	2003
18	17	527225035	50	RL	152	12134	Pave	NA	IR1	Bnk	AllPub	Inside	Mod	Gilbert	Norm	Norm	1fam	1.5Fin	8	7	1988
19	18	527258010	20	RL	88	11394	Pave	NA	Reg	Lvl	AllPub	Corner	Gtl	StoneBr	Norm	Norm	1fam	1story	9	2	2010
20	19	527276150	20	RL	140	19138	Pave	NA	Reg	Lvl	AllPub	Corner	Gtl	Gilbert	Norm	Norm	1fam	1story	4	5	1951
21	20	527302110	20	RL	85	13175	Pave	NA	Reg	Lvl	AllPub	Inside	Gtl	NWAmes	Norm	Norm	1fam	1story	6	6	1978
22	21	527358140	20	RL	105	11751	Pave	NA	IR1	Lvl	AllPub	Inside	Gtl	NWAmes	Norm	Norm	1fam	1story	6	6	1977
23	22	527358200	85	RL	85	10625	Pave	NA	Reg	Lvl	AllPub	Inside	Gtl	NWAmes	Norm	Norm	1fam	1story	7	6	1974
24	23	527388020	60	FV		7500	Pave	NA	Reg	Lvl	AllPub	Inside	Gtl	Somerst	Norm	Norm	1fam	2story	7	5	2000
25	24	527402200	20	RL		11241	Pave	NA	IR1	Lvl	AllPub	CulDSac	Gtl	NAmes	Norm	Norm	1fam	1story	6	7	1970

Figure 1: Dataset Overview

The Ames Housing dataset contains various property characteristics such as lot size, overall quality, neighborhood, garage information, and sale price. Data preprocessing was performed before building the prediction model.

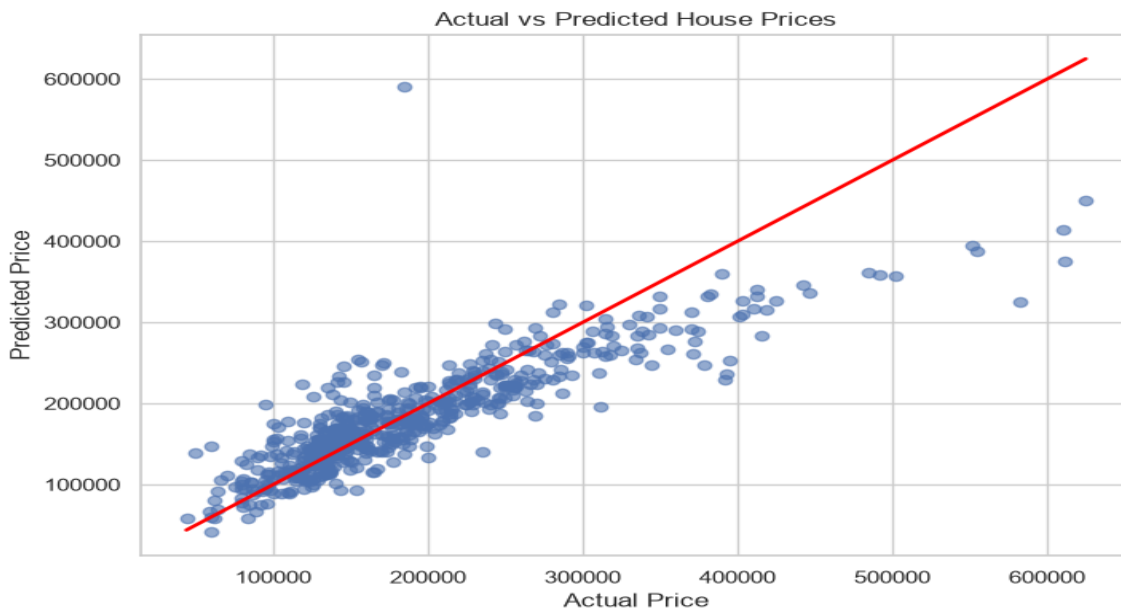


Figure 2: Actual vs Predicted

This plot compares the actual house prices with the predicted values generated by the Linear Regression model. Most points lie close to the diagonal reference line, indicating that the model predicts house prices reasonably well, although some prediction errors exist for expensive houses.

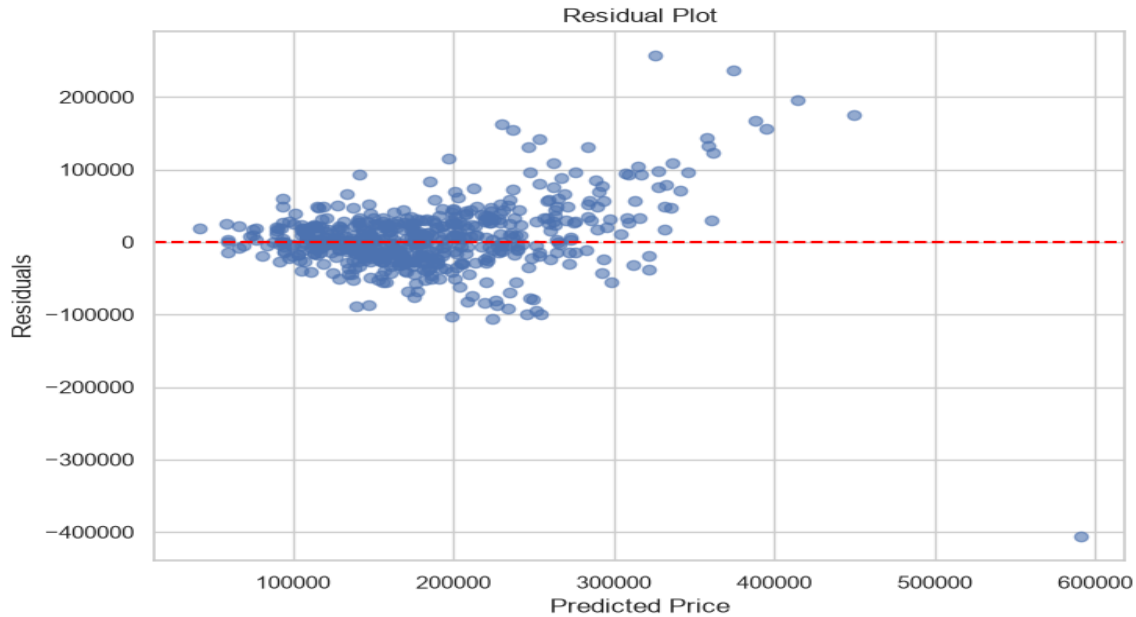


Figure 3: Residual Plot

The residual plot shows the prediction errors of the model. Most residuals are scattered around zero, indicating that the model performs reasonably well. However, larger errors appear for a few high-priced houses, suggesting that the model struggles slightly with extreme values.

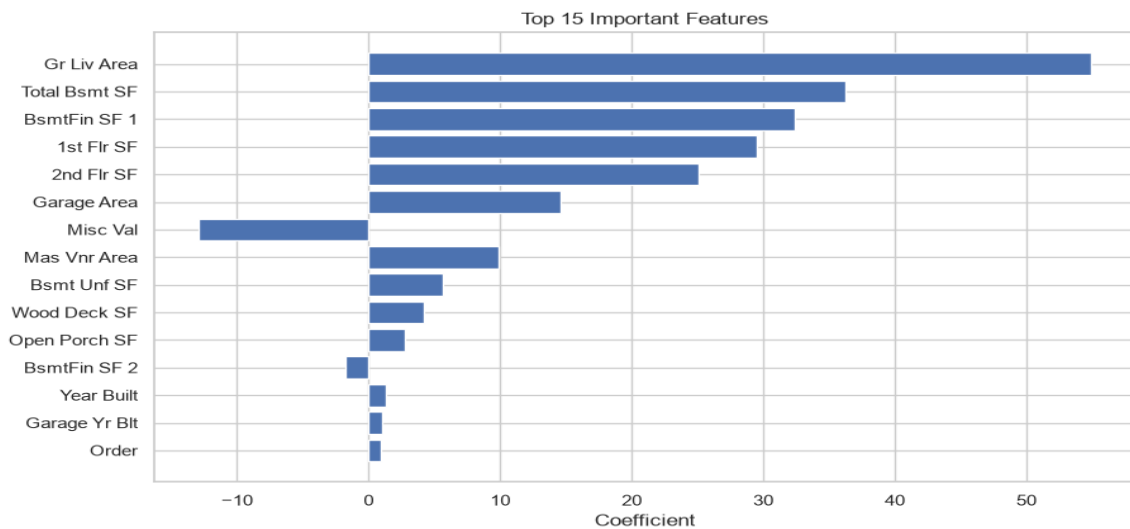


Figure 4: Top Feature Coefficients

This chart displays the most influential features used by the Linear Regression model. Features with larger positive coefficients contribute to increasing house prices, while negative coefficients reduce the predicted value. Living area and basement size have the strongest positive impact on sale price.

Key Findings

- Linear Regression achieved an **R² score of approximately 0.73**.

- Above-ground living area is the most influential feature.
- Basement size also strongly affects house price.
- The model predicts average-priced houses more accurately than expensive houses.
- Data preprocessing significantly improved model performance.

Business Recommendations

- Focus on improving living area when increasing property value.
- Renovate basements to enhance selling prices.
- Use additional machine learning models such as Random Forest or XGBoost for higher prediction accuracy.
- Continuously update the dataset with new housing information.
- Include more neighbourhood-specific features for better predictions.

Task Conclusion

The Linear Regression model successfully predicted house prices after preprocessing, handling missing values, and encoding categorical variables. With an R^2 score of approximately 0.73, the model demonstrates good predictive capability and provides valuable insights into the factors that influence residential property prices.

Skills Acquired

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook, Data Cleaning, EDA, Feature Engineering, K-Means Clustering, Linear Regression, Model Evaluation, Business Analytics.

Overall Conclusion

The internship provided practical exposure to solving real-world data analytics problems using Python. The projects covered exploratory analysis, clustering, and predictive modelling while strengthening analytical and documentation skills.